Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $x^m \cdot y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is

(A) $\frac{x+y}{xy}$

(B) xy

(C) $\frac{x}{y}$

(D) $\frac{y}{x}$

Q.2

If $x = a\cos^3 t, y = a\sin^3 t$, then $\frac{d^2y}{dx^2}$ is equal to

(A) $3a\sin^4 t\cos t$

(B) $\frac{1}{3a\cos^4 t\sin t}$

(C) $\frac{1}{3a\sin^4 t\cos t}$

(D) $3a\sec^4 t\sin t$

Q.3

If $\sin(x+y) = \log_e(x+y)$, then $\frac{dy}{dx}$ is equal to

(A) $\tan(x+y)$

(B) $\log_e(x+y)$

(C) -1

(D) 1

Q.4

If $\sin y = x\sin(a+y)$, then $\frac{dy}{dx}$ is

(A) $\frac{\sin y}{\sin a\sin^2(a+y)}$

(B) $\frac{\sin^2(a+y)}{\sin a}$

(C) $\sin a\sin^2(a+y)$

(D) none of these

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Q.5

If $y = x^y$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{y^2}{x(1+y\log_e x)}$

(B) $\frac{y^2}{x(1-y\log_e x)}$

(C) $\frac{y^2}{x^2(1-y\log_e x)}$

(D) $\frac{y^2}{x(1-\log_e x)}$

Q.6

If $x = a\left(t + \frac{1}{t}\right), y = a\left(t - \frac{1}{t}\right)$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{t^2-1}{t^2+1}$

(B) $\frac{1-t^2}{1+t^2}$

(C) $\frac{t^2+1}{1-t^2}$

(D) none of these

Q.7

If $\sin(xy) + \cos(xy) = 1$ and $\tan(xy) \neq 1$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{-y}{x}$

(B) $-\frac{x}{y}$

(C) $-xy$

(D) xy

Q.8

If $f(x) = \cot^{-1}\left(\frac{x^x - x^{-x}}{2}\right)$, then $f'(1)$ equals

(A) -1

(B) 1

(C) $\log 2$

(D) $-\log 2$

Q.9

Let $f(x) = \frac{x^2}{1-x^2}, x \neq 0, \pm 1$, then the derivative of $f(x)$ with respect to x^2 is

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(A) $\frac{2x}{(1-x^2)^2}$

(B) $\frac{1}{(1-x^2)^2}$

(C) $\frac{1}{(2+x^2)^2}$

(D) $\frac{1}{(2-x^2)^2}$

Q.10

If $f(x) = \log_a \log_a(x)$, then $f'(x)$ is

(A) $\frac{\log_a e}{x \log_e x}$

(B) $\frac{\log_e a}{x \log_a x}$

(C) $\frac{\log_e a}{x}$

(D) $\frac{x}{\log_e a}$

Q.11

If $y = \sqrt{x + \sqrt{y + \sqrt{x + \ldots \infty}}}$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{1}{2y-1}$

(B) $\frac{y^2-x}{2y^3-2xy-1}$

(C) 2y-1

(D) none of these

Q.12

The second derivative of $a \sin^3 t$ with respect to $a \cos^3 t$ at $t = \frac{\pi}{4}$ is

(A) $\frac{4\sqrt{2}}{3a}$

(B) 2

(C) $\frac{1}{12a}$

(D) none of these

Q.13

$\frac{d}{dx}\left(\tan^{-1}\sqrt{\frac{1+\cos\frac{x}{2}}{1-\cos\frac{x}{2}}}\right)$ is equal to

(A) $-\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

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(D) $\frac{1}{4}$

Q.14

A curve is given by the equations $x = a\cos\theta + \frac{1}{2}b\cos 2\theta, y = a\sin\theta + \frac{1}{2}b\sin 2\theta$, then the $\frac{d^2y}{dx^2} = 0$ points for which dx^2 are given by

(A) $\sin\theta = \frac{2a^2+b^2}{5ab}$

(B) $\tan\theta = \frac{3a^2+2b^2}{4ab}$

(C) $\cos\theta = \frac{-\left(a^2+2b^2\right)}{3ab}$

(D) $\cos\theta = \frac{\left(a^2-2b^2\right)}{3ab}$

Q.15

If $\cos^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right) = \log a$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{y}{x}$

(B) $\frac{x}{y}$

(C) $\frac{x^2}{y^2}$

(D) $\frac{y^2}{x^2}$

Q.16

Let $f(x)$ be a quadratic expression which is positive for all x. If $g(x) = f(x) + f'(x) + f''(x)$, then for all real x

(A) $g(x) < 0$

(B) $g(x) > 0$

(C) $g(x) = 0$

(D) $g(x) \geq 0$

Q.17

If $f(x) = \log_{x^2}(\ln x)$, then $f'(x)$ at $x = e$, is

(A) 0

(B) 1

(C) $1/e$

(D) $1/(2e)$

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Q.18

If $y^2 = P(x)$, a polynomial of degree 3, then $2\frac{d}{dx}\left(y^3 \frac{d^2y}{dx^2}\right)$ equals

(A) $P'''(x) + P'(x)$

(B) $P''(x) + P'''(x)$

(C) $P(x) \cdot P'''(x)$

(D) a constant

Q.19

If $y = \log_e\left(\sec\left(e^{x^2}\right)\right)$, then $\frac{dy}{dx}$ is equal to

(A) $2x\left(\tan e^{x^2}\right)e^{x^2}$

(B) $2x\left(e^{x^2}\left(\sec e^{x^2}\right)\tan e^{x^2}\right)$

(C) $x^2 e^{x^2}\tan e^{x^2}$

(D) $e^{x^2}\tan e^{x^2}$

Q.20

If $y^2 = ax^2 + bx + c$, then $y^3\frac{d^2y}{dx^2}$ is

(A) a constant

(B) a function of x only

(C) a function of y only

(D) a function of x and y

Q.21

If $xe^{xy} = y + \sin^2 x$, then at $x = 0$, $\frac{dy}{dx}$ is equal to

Q.22

Then $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a constant. Then $\frac{d^3}{dx^3(f(x))}$ at $x = 0$ is

Q.23

If $y = \frac{\cos 6x + 6\cos 4x + 15\cos 2x + 10}{\cos 5x + 5\cos 3x + 10\cos x}$, then $\frac{dy}{dx}$ is equal to $-\frac{n}{4}\sin x$ then $n =$

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Q.24

If $\frac{d}{dx}(f(x)) = e^{-x}f(x) + e^{x}f(-x)$, then $f(10) + f(-10) =$.(given $f(0) = 0$)

Q.25

Let f be a function defined for all $x \in R$. If f is differentiable and $f\left(x^3\right) = x^5$ for all $x \in R(x \neq 0)$, then the value of $f'(27)$ is

Q.26

The derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ is $(-1 < x < 1)$

Q.27

If $5f(x) + 3f\left(\frac{1}{x}\right) = x + 2$ and $y = xf(x)$, then $\left(\frac{dy}{dx}\right)$ at $x = 1$ is equal to (mark answer to nearest integer)

Q.28

If $y = (1 + x)\left(1 + x^2\right)\left(1 + x^4\right) \ldots . \left(1 + x^{2n}\right)$, then $\frac{dy}{dx}$ at $x = 0$ is

Q.29

If $y = \sqrt{\sin x + y}$ then $\frac{dy}{dx}$ at $(0, 1)$ is equal to

Q.30

If $f(x + y) = f(x) \cdot f(y)$ for all $x, y \in R$ and $f(5) = 2, f'(0) = 3, f'(5) =$

Answers & Solutions

Q.1 Answer:

$\frac{y}{x}$

Solution:

$$x^m \cdot y^n = (x + y)^{m+n} \Rightarrow m \log x + n \log y = (m + n) \log(x + y)$$
$$\Rightarrow \frac{m}{x} + \frac{n}{y} \cdot \frac{dy}{dx} = \frac{m+n}{x+y}\left(1 + \frac{dy}{dx}\right) \Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

Q.2 Answer:

$\frac{1}{3a\cos^4 t \sin t}$

Solution:

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$\frac{dx}{dt} = -3a\cos^2 t \sin t, \frac{dy}{dt} = 3a\sin^2 t \cos t$
$\frac{dy}{dx} = -\tan t, \frac{d^2y}{dx^2} = \frac{d}{dt}(-\tan t)\cdot\left(\frac{dt}{dx}\right) = -\sec^2 t \cdot \left(-\frac{1}{3a\cos^2 t \sin t}\right) = \frac{1}{3a\cos^4 t \sin t}$

Q.3 Answer:

-1

Solution:

$\sin(x+y) = \log_e(x+y) \Rightarrow \cos(x+y)\left(1+\frac{dy}{dx}\right) = \frac{1}{x+y}\cdot\left(1+\frac{dy}{dx}\right) \Rightarrow 1+\frac{dy}{dx} = 0$
$\Rightarrow \quad \frac{dy}{dx} = -1$

Q.4 Answer:

$\frac{\sin^2(a+y)}{\sin a}$

Solution:

$\sin y = x\sin(a+y)$ **or** $x = \frac{\sin y}{\sin(a+y)}$
$\Rightarrow \quad \frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y\cos(a+y)}{\sin^2(a+y)}$
$\Rightarrow \frac{dx}{dy} = \frac{\sin(a+y-y)}{\sin^2(a+y)} \Rightarrow \frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$

Q.5 Answer:

$\frac{y^2}{x(1-y\log_e x)}$

Solution:

$\log_e y = y\log_e x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \log_e x\cdot\frac{dy}{dx} + \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1-y\log_e x)}$

Q.6 Answer:

none of these

Solution:

$\frac{dx}{dt} = a\left(1-\frac{1}{t^2}\right) = a\left(\frac{t^2-1}{t^2}\right)$
$\frac{dy}{dt} = a\left(1+\frac{1}{t^2}\right) = a\left(\frac{t^2+1}{t^2}\right), \quad \therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{t^2+1}{t^2-1}$

Q.7 Answer:

$\frac{-y}{x}$

Solution:

$\cos(xy)\left\{y + x\frac{dy}{dx}\right\} - \sin(xy)\left\{y + x\frac{dy}{dx}\right\} = 0$
$\left(y + x\frac{dy}{dx}\right)(\cos(xy) - \sin(xy)) = 0 \quad y + x\frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{-y}{x}$

Q.8 Answer:

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-1

Solution:

Putting $u = x^x$ and using logarithmic differentiation, we get $\log u = x \log x$.

$\Rightarrow \quad \frac{1}{u}\frac{du}{dx} = x \cdot \frac{1}{x} + \log x \Rightarrow \frac{du}{dx} = x^x(1 + \log x)$

Similarly, if $v = x^{-x}$, we get, $\frac{dv}{dx} = -x^{-x}(1 + \log x)$

$\therefore f'(x) = -\frac{1}{1+\left(\frac{x^x - x^{-x}}{2}\right)^2}\frac{d}{dx}\left(\frac{x^x - x^{-x}}{2}\right)$

$= -\frac{4}{x^{2x}+x^{-2x}+2}\left[\frac{1}{2}\left[\left(x^x + x^{-x}\right)(1 + \log x)\right]\right] \Rightarrow f'(1) = -\frac{2}{1+1+2}[2(1 + \log 1)] = -1$

Q.9 Answer:

$\frac{1}{(1-x^2)^2}$

Solution:

$f(x) = \frac{x^2}{1-x^2} \Rightarrow \quad \frac{d}{dx}(f(x)) = \frac{\left(1-x^2\right)2x - x^2(-2x)}{(1-x^2)^2} = \frac{2x(1)}{(1-x^2)^2}$

Also, $\frac{d}{dx}\left(x^2\right) = 2x$, therefore $\frac{df(x)}{d(x^2)} = \frac{1}{(1-x^2)^2}$

Q.10 Answer:

$\frac{\log_a e}{x \log_e x}$

Solution:

$f'(x) = \frac{1}{\log_a(x)} \cdot \log_a e \frac{d}{dx}\left[\log_a x\right]$

$= \frac{1}{\log_a x} \cdot \log_a e \cdot \frac{1}{x} \cdot \log_a e = \frac{\log_a e}{x}\frac{\log_a e}{\log_a x} = \frac{\log_a e}{x \log_e x}$

Q.11 Answer:

$\frac{y^2-x}{2y^3-2xy-1}$

Solution:

$y = \sqrt{x + \sqrt{y + \sqrt{x + \ldots \infty}}} \Rightarrow y = \sqrt{x + \sqrt{y + y}} \Rightarrow y = \sqrt{x + \sqrt{2y}}$

$\Rightarrow \left(y^2 - x\right)^2 = 2y \Rightarrow 2\left(y^2 - x\right)\left(2y\frac{dy}{dx} - 1\right) = 2\frac{dy}{dx}$

$\Rightarrow 2y\left(y^2 - x\right)\frac{dy}{dx} - \frac{dy}{dx} = y^2 - x \Rightarrow \frac{dy}{dx} = \frac{y^2-x}{2y^3-2xy-1}$

Q.12 Answer:

$\frac{4\sqrt{2}}{3a}$

Solution:

Let $y = a\sin^3 t$ and $x = a\cos^3 t$, then

On differentiating with respect to t, we get $\frac{dy}{dt} = 3a\sin^2 t\cos t$

and $\frac{dx}{dt} = 3a\cos^2 t(-\sin t)$

$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{3a\sin^2 t\cos t}{3a\cos^2 t(-\sin t)} = -\tan t$

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Again differentiating with respect to x

$\frac{d^2y}{dx^2} = -\sec^2 t \frac{dt}{dx} = \frac{-\sec^2 t}{3a\cos^2 t(-\sin t)} = \frac{1}{3a}\left(\frac{\sec^4 t}{\sin t}\right)$

$\therefore \left(\frac{d^2y}{dx^2}\right)_{t=\frac{\pi}{4}} = \frac{1}{3a} \cdot \frac{4}{\frac{1}{\sqrt{2}}} = \frac{4\sqrt{2}}{3a}$

Q.13 Answer:

$-\frac{1}{4}$

Solution:

$y = \tan^{-1}\sqrt{\frac{1+\cos x/2}{1-\cos x/2}} = \tan^{-1}\sqrt{\frac{2\cos^2 x/4}{2\sin^2 x/4}}$

Let $y = \tan^{-1}\cot\frac{x}{4} = \tan^{-1}\tan\left(\frac{\pi}{2} - \frac{x}{4}\right) = \frac{\pi}{2} - \frac{x}{4}; \therefore \frac{dy}{dx} = -\frac{1}{4}$

Q.14 Answer:

$\cos\theta = \frac{-\left(a^2+2b^2\right)}{3ab}$

Solution:

$x = a\cos\theta + \frac{1}{2}b\cos 2\theta; y = a\sin\theta + \frac{1}{2}b\sin 2\theta$

$\frac{dy}{d\theta} = a\cos\theta + b\cos 2\theta, \frac{dx}{d\theta} = -a\sin\theta - b\sin 2\theta$

$\therefore \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a\cos\theta+b\cos 2\theta}{-a\sin\theta-b\sin 2\theta}$

$\therefore \quad \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{d\theta}\cdot\left(\frac{dy}{dx}\right)\cdot\frac{d\theta}{dx}$

$= \left[\frac{(a\sin\theta+b\sin 2\theta)(a\sin\theta+2b\sin 2\theta)}{(a\sin\theta+b\sin 2\theta)^2} + \frac{(a\cos\theta+b\cos 2\theta)(a\cos\theta+2b\cos 2\theta)}{(a\sin\theta+b\sin 2\theta)^2}\right]\cdot\frac{d\theta}{dx}$

but $\frac{d^2y}{dx^2} = 0$

$\Rightarrow \quad a^2 + 2b^2 + 3ab[\sin 2\theta\sin\theta + \cos 2\theta\cdot\cos\theta] = 0$

$\Rightarrow a^2 + 2b^2 = -3ab\cos(2\theta - \theta)$

$\therefore \quad \cos\theta = -\left(\frac{a^2+2b^2}{3ab}\right)$

Q.15 Answer:

$\frac{y}{x}$

Solution:

$\cos^{-1}\left(\frac{x^2-y^2}{x^2+y^2}\right) = \log a \quad \frac{x^2-y^2}{x^2+y^2} = \cos(\log a)$

$\Rightarrow \frac{x^2}{y^2} = \textbf{constant} \Rightarrow \frac{x}{y} = \textbf{constant} \Rightarrow \quad \frac{y - x\cdot\frac{dy}{dx}}{y^2} = 0$

$\Rightarrow \quad y - x\cdot\frac{dy}{dx} = 0 \quad \Rightarrow \frac{dy}{dx} = \frac{y}{x}$

Q.16 Answer:

$g(x) > 0$

Solution:

Let $f(x) = ax^2 + bx + c$

As $f(x) > 0, \forall x \in R$**, we have,**

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$a > 0$ and $b^2 - 4ac < 0$

Also, $f'(x) = 2ax + b$ and $f''(x) = 2a$

Thus, $g(x) = ax^2 + (2a + b)x + 2a + b + c$

Since $a > 0$ and discriminant is

$\therefore (2a + b)^2 - 4a(2a + b + c)$

$= 4a^2 + 4ab + b^2 - 8a^2 - 4ab - 4ac$

$= -4a^2 + \left(b^2 - 4ac\right) < 0 \left[\because b^2 - 4ac < 0\right]$

$g(x) > 0, \forall x > 0$

Q.17 Answer:

$1/(2e)$

Solution:

$y = f(x) = \frac{\log_e(\log_e x)}{2\log_e x} = \frac{\log_e t}{2t}$

We have $2\log_e x \quad 2t$, where $t = \log_e x$.

$\therefore f'(x) = \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{2} \cdot \frac{(1/t)\cdot t - 1\cdot \log_e t}{t^2} \cdot \frac{1}{x}$

Now at $x = e$, we have $t = \log_e e = 1$

at $x = e, \quad f'(x) = \frac{1}{2} \cdot \frac{1 - \log_e 1}{1^2} \cdot \frac{1}{e} = \frac{1}{2e}$

Q.18 Answer:

$P(x) \cdot P'''(x)$

Solution:

We have $P'(x) = 2yy', P''(x) = 2y'^2 + 2yy''$

and $\quad P'''(x) = 6y'y'' + 2yy'''$

$2\frac{d}{dx}\left(y^3 \frac{d^2y}{dx^2}\right) = 2\frac{d}{dx}\left(y^3 y''\right)$

Now $= 2\left(y^3 y''' + 3y^2 y' y''\right) = y^2\left(2yy''' + 6y'y''\right) = P(x)P'''(x)$

Q.19 Answer:

$2x\left(\tan e^{x^2}\right)e^{x^2}$

Solution:

Let $y = \log\left[\sec\left(e^{x^2}\right)\right]$

Differentiating w.r.t. x,

we get $\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{d}}{\mathrm{dx}} \log\left[\sec\left(e^{x^2}\right)\right]$

$= \frac{1}{\sec\left(e^{x^2}\right)} \cdot \frac{\mathrm{d}}{\mathrm{dx}}\left[\sec\left(e^{x^2}\right)\right]$

$= \frac{1}{\sec\left(e^{x^2}\right)} \cdot \sec\left(e^{x^2}\right)\tan\left(e^{x^2}\right) \cdot \frac{\mathrm{d}}{\mathrm{dx}}\left(e^{x^2}\right)$

$= \tan\left(e^{x^2}\right) \cdot e^{x^2} \cdot \frac{d}{dx}\left(x^2\right)$

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$= \tan\left(e^{x^2}\right) \cdot e^{x^2} \cdot 2x$

$= 2x \cdot e^{x^2} \tan\left(e^{x^2}\right)$

Q.20 Answer:

a constant

Solution:

Given that $y^2 = ax^2 + bx + c$

$2y \cdot \frac{dy}{dx} = 2ax + b$ **...(i)**

$2\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] = 2a \quad y \cdot \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = a$ **...(ii)**

(i) and (ii) $\frac{yd^2y}{dx^2} = a - \left(\frac{2ax+b}{2y}\right)^2$ **put** $y^2 = ax^2 + bx + c$

$\Rightarrow \quad y^3\frac{d^2y}{dx^2} = \frac{1}{4}\left(4a\left(ax^2 + bx + c\right) - \left(4a^2x^2 + b^2 + 4abx\right)\right)$

$= \frac{1}{4}\left(4a^2x^2 + 4abx + 4ac - 4a^2x^2 - b^2 - 4abx\right) = \frac{1}{4}\left(4ac - b^2\right)$

Q.21 Answer:

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Solution:

$xe^{xy} = y + \sin^2 x \Rightarrow e^{xy} + xe^{xy} \cdot \left(x\frac{dy}{dx} + y \cdot 1\right) = \frac{dy}{dx} + 2\sin x \cos x$

$e^0 + 0 = \frac{dy}{dx} + 0 \quad \Rightarrow \frac{dy}{dx} = 1$

Q.22 Answer:

—

Solution:

$f'(x) = \begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}; \quad f''(x) = \begin{vmatrix} 6x & -\sin x & -\cos x \\ 5 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

$f'''(x) = \begin{vmatrix} 6 & -\cos x & \sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}; \quad f'''(x) = \begin{vmatrix} 6 & -1 & 0 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix} = 0$

$[QR_1, R_2$ **are identical]**

Q.23 Answer:

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Solution:

Numerator $= \cos 6x + \cos 4x + 5(\cos 4x + \cos 2x) + 10(\cos 2x + 1)$

$= 2\cos x\{\cos 5x + 5\cos 3x + 10\cos x\}$

$\therefore y = \frac{\text{Nr}}{\text{Dr}} = 2\cos x, \frac{dy}{dx} = -2\sin x$

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Q.24 Answer:

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Solution:

Let $g(x)=\frac{d}{dx}(f(x))$; Now $g(-x)=g(x)$

$\therefore$ **Since derivative of $f(x)$ is an even function therefore $f(x)$ must be an odd function.**

Q.25 Answer:

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Solution:

We have, for all $x(x\neq 0)\Rightarrow \quad f\left(x^3\right)=x^5$

Differentiating w.r.t. x**, we get** $f'\left(x^3\right)\cdot 3x^2=5x^4\Rightarrow f'\left(x^3\right)=\frac{5}{3}x^2$

$\therefore f'(27)=f'\left(3^3\right)=\frac{5}{3}(3)^2=15$

Q.26 Answer:

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Solution:

Let $y_1=\tan^{-1}\frac{2x}{1-x^2}$

and $y_2=\sin^{-1}\frac{2x}{1+x^2}$.

Differentiating y_1 **and** y_2 **with respect to** x**, we get**

$\frac{dy_1}{dx}=\frac{d}{dx}\left(\tan^{-1}\frac{2x}{1-x^2}\right)$ **(1)**

$\frac{dy_2}{dx}=\frac{d}{dx}\left(\sin^{-1}\frac{2x}{1+x^2}\right)$ **(2)**

Putting $x=\tan\theta$**, we have** $y_1=\tan^{-1}\tan 2\theta=2\theta=2\tan^{-1}x$

and $y_2=\sin^{-1}\sin 2\theta=2\tan^{-1}x$

Again $\frac{dy_1}{dx}=\frac{d}{dx}\left(2\tan^{-1}x\right)=\frac{2}{1+x^2}$

and $\frac{dy_2}{dx}=\frac{d}{dx}\left(2\tan^{-1}x\right)=\frac{2}{1+x^2}$

Hence,

$\frac{dy_1}{dy_2}=1$

Q.27 Answer:

—

Solution:

$5\mathrm{f(x)}+3\mathrm{f}\left(\frac{1}{\mathrm{x}}\right)=\mathrm{x}+2---(1)$

so sf $\left(\frac{1}{x}\right)+3(x)=\frac{1}{x}+2----(2)$

Multiply $5(1)\&3(2)\&$ **subtract thems**

$\frac{25\mathrm{f(x)}+15\mathrm{f}\left(\frac{1}{\mathrm{x}}\right)=5(\mathrm{x}+2)-9\mathrm{f(x)}+15\mathrm{f}\left(\frac{1}{\mathrm{x}}\right)=3\left(\frac{1}{\mathrm{x}}+2\right)}{16\mathrm{f(x)}=5\mathrm{x}+10-\frac{3}{\mathrm{x}}-6\qquad \mathrm{f(x)}=\frac{1}{16}\left(5\mathrm{x}+4-\frac{3}{\mathrm{x}}\right)}$

$y=xf(x)$

$=x\left(\frac{1}{16}\left(5x+4-\frac{3}{x}\right)\right)$

$\frac{1}{16}\left(5x^2+4x-3\right)$

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$\left.\frac{dy}{dx}\right|_{x-1} \Rightarrow \frac{1}{16}(10x+4)$
$= \frac{1}{16}(10(1)+4)$
$= \frac{14}{16}$
$= 1$

Q.28 Answer:

—

Solution:

We have, $y = (1+x)\left(1+x^2\right)\left(1+x^4\right)\ldots\ldots\left(1+x^4\right)$
take natural logarithm both sides $\ln y = \ln(1+x) + \ln\left(1+x^2\right) + \ln\left(1+x^4\right) + \ldots\ldots\ldots\ldots\ldots + \ln\left(1+x^{2n}\right)$
Now differentiate both sides w.r.t. $x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \ldots.. + \frac{2nx^{2n-1}}{1+x^{2x}}$
$\Rightarrow \frac{dy}{dx} = y\left(\frac{1}{x} + \frac{2x}{1+x^2} + \ldots\ldots\ldots + \frac{2nx^{2n-1}}{1+x^{2n}}\right)$
Now substitute $x = 0$ **to get the required value** $\Rightarrow \left.\frac{dy}{dx}\right|_{x=0} = 1(1+0+0+\ldots.. + 0) = 1$
Note that at $x = 0$**, value of** y **is also 1**

Q.29 Answer:

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Solution:

Since $y = \sqrt{\sin x + y} \quad [\sqrt{f(x)+y}]$
$\therefore \frac{dy}{dx} = \frac{\frac{d}{dx}(\sin x)}{2y-1} = \frac{\cos x}{2y-1}$

Q.30 Answer:

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Solution:

$f(x+h) = f(x)f(h)$
Putting $x = 0, y = 5$ **in the given equation, we get**
$f(0+5) = f(0)f(5)$
$\Longrightarrow f(5)[f(0)-1] = 0$
$\Longrightarrow f(0) = 1$
Consider, $f'(5) = \lim_{h\to 0}\frac{f(5+h)-f(5)}{h}$
$= \lim_{L\to 0}\frac{f(5)f(h)-f(5)}{h}$
$= \lim_{L\to 0}\frac{f(5)f(h)-1]}{h}$
$= f(5)\lim_{h\to 0}\frac{f(h)-f(0)}{h}$
$= f(5)f'(0) = 2\times 3 = 6$
$\therefore f'(5) = 6$

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